README

This document contains instructions and notes for the technical set-up the GRIP experiment for the UCBED study at the Shriners. The study has two sessions: a morning session where an ultrasound experiment and an EMG experiment will be, and an afternoon session where the same task will be performed while participants lie in an fMRI machine at the Imaging Reseach Center in Sacramento. All data are collected in Sacramento.

Any notes on communication with participants is not included here.

Directories Shorthand

grip	C:\Users\ucdbe\grip
simultaneous emg ultrasound	C:\Users\ucdbe\grip\Task_Peripheral\simultaneous-emg-ultrasound
ultrasound	C:\Users\ucdbe\grip\Task_Peripheral\ultrasound
EMG	C:\Users\ucdbe\grip\Task_Peripheral\emg
Ultrasound subject-data	C:\Users\ucdbe\grip\Task_Peripheral\ultrasound\subject-data
EMG subject-data	C:\Users\ucdbe\grip\Task_Peripheral\emg\subject-data
Stim presentation	C:\Users\ucdbe\grip\Task_Peripheral\stim_presentation
Conditions files	C:\Users\ucdbe\grip\Task_Peripheral\stim_presentation\conditions

Contents:

Instructions for different cases are below:

1. Instructions for Simultaneous EMG and Ultrasound

Before Starting Data Collection:

1. Plug in and set up EMG system, ensuring electrodes are charged, connecting the DAQ, and connect both wires from Delsys and DAQ to the PC, and attaching stickers etc.
2. Set up and start the two ultrasound machines and connect them to the computer.
3. Connect the webcam to the computer.
4. Put electrodes on participant's arm in the part of the arm with the biggest bulge when they flex on the volar surface. (1-8 on affected arm, 9-16 on unaffected arm).
5. Open EMGWorks and check signal from electrodes. Close EMGWorks.
6. Open Trigno Utility and verify sensors are connected.
7. In MATLAB, in the EMG directory, run Delsys_DAQ_Testing.m and confirm signal quality. Close plots.
8. Close / quit MATLAB. Then reopen (to prevent an error related to DAQ).
9. Secure ultrasound probes and ensure all machines are connected to the PC.
10. Follow the instructions in the "For Each Run" section using subject name **'test'** so all programs are open and ready before the participant arrives. Set up the first run as described below.

For Each Run:

1. Launch **LabRecorder.exe** from the Start Menu.
2. In LabRecorder:
    - Go to **File → Load Configuration**
    - Choose the configuration file in the simultaneous ultrasound emg directory.
    - Once loaded, change the following:
        - Block/Task: change Default to grip
        - Update Run
        - Update Participant
        - Update Session
        - Select beh for Modality
3. Launch **PsychoPy Builder**.
    - Open grip_task.psyexp
    - Switch the toggle from **Pilot** to **Run**
4. Move LabRecorder to the second monitor.
5. Open MATLAB.

6. Navigate to the directory containing the simultaneous recording script.
7. Open:

   recordSimultaneousEmgUltrasound.m

8. Verify camera assignments:
   - uscam = NEW Terason (affected/missing hand)
   - uscam2 = OLD Terason (typical hand)
   - rwcam = webcam
9. Move MATLAB to the second monitor.
10. In PsychoPy:
    - Press the green **Run** button.
    - Click OK.
    - Enter participant ID.
    - Confirm mode is run.
    - Set paradigm appropriately for the simultaneous experiment. You can choose emg or ultrasound – this is where the conditions file will be read from. Just log whatever you choose in the run sheet. Be consistent. Choose either emg or ultrasound for all runs with a given subject.
    - Confirm and record the order of the conditions file in run sheet.
11. Run:

    recordSimultaneousEmgUltrasound.m

12. Enter:
    - Subject ID
    - Run number
13. Wait for MATLAB to connect to the LSL marker stream.

    NOTE: If MATLAB times out waiting for LSL, PsychoPy was not started correctly. Close MATLAB and restart PsychoPy first.

14. MATLAB will:
    - Connect to Delsys
    - Initialize cameras
    - Initialize LSL
    - Warm up the parallel workers
15. MATLAB will ask:

    "Did you press Start Recording in LabRecorder?"

    Before answering:

    - Confirm all required streams are green in LabRecorder.

- Press **Update** if necessary.
- Press **Start Recording**.

16. Continue answering MATLAB prompts until it reports that setup is complete and it is okay to proceed.
17. Move the cursor to the main monitor.
18. Press any key to skip instructions.
19. Press:

    ```
    t
    ```

    to trigger the PsychoPy experiment.

20. At the end of the run:
    - Press **Stop** in LabRecorder.

NOTE: At the End of a Successful Run

You should see:

MATLAB files

Inside:

```
subject-data/<SubjectID>/<RunNumber>
```

Files for each:

- grasp
- repeat
- ultrasound camera 1
- ultrasound camera 2
- real-world camera
- EMG recording

For 5 grasps, 5 repeats, you should have 4 kinds of files: EMG, RW, US and US2 across two limbs.

XDF file

Inside:

```
subject-data/sub-<SubjectID>/
```

An .xdf file containing timing markers and synchronization streams.

Recommended Checks After Run 1

1. Spot-check ultrasound movies using:
   i. viewMovie(filepath)
2. Verify:
   a. Ultrasound movies show motion.
   b. Real-world video is not blank.
   c. Timing appears correct.
3. Open one EMG file and confirm signal is present and not flat.
4. Load the XDF file (you may need to add xdf-Matlab in Task_Peripheral to the path)
   i. timings = load_xdf(filepath_to_xdf);
5. Confirm the expected streams are present and non-empty.

At the end of the full experiment:

1. Take pictures of the set up/ location of probes on participant's arms.
2. Disconnect apparatus from participant.
3. Take measurements:
   a. limb circumference(s) at the widest part of arm(s)
   b. angle of extension
   c. angle of flexion
   d. limb length (to wrist of unaffected arm, to end of residuum)
4. Shut down and pack up equipment.

2. Instructions for Ultrasound Experiment

Before Starting Data Collection:

1. Set up and start the two ultrasound machines and connect them to the computer.
2. Connect the webcam to the computer.
3. Follow the instructions in the 'For each run' section with subject name 'test' so all the programs are open and ready.
4. Connect participant to equipment.
5. Set up the first run (as below).

For Each Run:

1. Launch LabRecorder.exe from the Start Menu.
2. In LabRecorder, go to **File**, then **Load Configuration**.
3. Choose the cfg file in ultrasound directory (see 'Directories Shorthand' table).

4. Once loaded change the following:
    a. In **Block/Task**, change ‘Default’ to ‘ultrasound-grip’
    b. Update **Run** to whichever run of PsychoPy this corresponds to. Leave as 1 if your first run.
    c. Update **Participant**.
    d. Update **Session.** This will all be in session 1. Change to 01 instead of S001.
    e. Select ‘beh’ for **Modality.**
5. Next, Launch **PsychoPy Builder** from Start Menu.
6. In PsychoPy Builder, go to **File**, then **Open**, then navigate to the stim presentation directory and select the **grip_task.psyexp** file.
   NOTE: Never edit grip_task_lastrun.py or it can corrupt the psyexp file.
7. In PsychoPy Builder, switch the toggle in the top bar from **Pilot** to **Run**.
8. At this point, you should already have a second monitor connected.
9. Move LabRecorder over to the second monitor.
   (See ‘Running With One Screen’ below, if that is not possible – not ideal. I would only suggest it if during a session, the extra monitor breaks and nothing else is possible.)
10. Open MATLAB.
11. Navigate to the ultrasound subdirectory (in UCBED directory).
12. Open the script recordUltrasoundDataSyncWithPsychoPy.m.
13. **Update lines 6, 7 and 8** in script to point to correct cameras: uscam for NEW Terason (for missing hand), uscam2 for OLD Terason (for typical arm), rwcam for webcam.
14. Move MATLAB window over to the second monitor.
15. In PsychoPy, press the green **Run** button to start the experiment. Hit **OK**.
16. In the modal that pops up, input the participant (same ID as you entered in LabRecorder). Check that mode is ‘run.’
    NOTE: The other option for ‘mode’ is ‘test’ and the difference is that the test mode does not have the extra 15 seconds of wait time before the first prompt appears. It’s okay to keep it in ‘run’ mode for all runs, for consistency and practice.
17. In **paradigm**, update fmri to ultrasound.
18. Ensure the conditions file number is correct. If first run, it’s okay. If not first run, change to the appropriate filename. Ensure you are reading the conditions file for the ultrasound exp not EMG experiment.
19. Run the MATLAB script recordUltrasoundDataSyncWithPsychoPy.m and type in participant ID (same as in PsychoPy and LabRecorder), for **Subject ID.**
20. Enter the **run number** (same as in PsychoPy and LabRecorder).
21. Run, and wait for LSL inlet to be connected. If this times out, it means the PsychoPy task was not on.

22. MATLAB will remind you to check for two green streams (one from Psychopy, one from MATLAB) in LabRecorder, and to press '**Start.**' If you don't see them, you may need to press the **Update** button first. Start Recording.
23. Continue to answer the MATLAB prompts till it says that it is okay to proceed.
24. Move your cursor to the main monitor, and press any key to skip instructions. Press '**t'** to trigger the PsychoPy experiment.
25. At the end of the run, press **Stop** on LabRecorder.

    NOTE: At the end of a successful run, you should see:

    - in **subject-data/subID:** a file for each cam, repeat and grasp (5 grasps × 2 limbs × 3 cams × 5 repeats = 150 mat files).
    - in **subject-data/sub-<SubID>/…**, an xdf file (with timing markers).

Recommended Checks After Run 1:
These can be done before the participant arrives with a brief test run on a lab member to check that everything works, if you'd rather save time.

a. Spot-check any US, US2 and RW movie using the helper function 'viewMovie(filepath)' (in 'Task_Peripheral' folder). It can be passed in the path to the mat file you want to view. Check to see the ultrasound shows movement, and the RW movie shows a full grasp, i.e., not blank, not stalled, and not misaligned in time.
b. Load the LSL (.xdf) file, with the function load_xdf(filepath_to_xdf) (make sure xdf-Matlab package in the Task_Peripheral folder is added to path).

   It should contain two non-empty structs (one for each stream).
   timings = load_xdf(filepath_xdf);
   timings{1}.time_series'
   timings{2}.time_series'

At the end of the full Ultrasound Experiment:

5. Take pictures of the set up/ location of probes on participant's arms.
6. Disconnect apparatus from participant.
7. Take measurements:
   a. limb circumference(s) at the widest part of arm(s)
   b. angle of extension
   c. angle of flexion

    d. limb length (to wrist of unaffected arm, to end of residuum)
8. Shut down and pack up equipment.
9. Check that files were saved where you expect them to be saved.
10. Check roughly that there are the expected number of files (for 10 runs, 5 grasps per run times 5 repeats per run times 3 types of files times 2 limbs = 10.5.5.3.2 = 1500 files.
11. Spot-check a couple files with viewMovie.m before EMG session.

    e.g., for one trial, view the US and US2 movies. There should be visible motion in one movie but not the other (if the participant was moving only one hand).

3. Instructions for Starting EMG Experiment

Before Starting Data Collection:

1. Set up the EMG system, charge electrodes, attach stickers, connect DAQ, connect both wires from Delsys and DAQ to the PC.
2. Connect the webcam to the computer.
3. Put electrodes on participant's arm in the part of the arm with the biggest bulge, when they flex on the volar surface. (1-8 on affected arm, 9-16 on unaffected arm).
4. Open EMGWorks, and check signal from electrodes.
5. Close EMGWorks.
6. Open Trigno Utility.
7. Run Trigno Utility.
8. In MATLAB, in emg directory (see Directory Shorthand table), run script Delsys_DAQ_Testing.m. Close plots.

For Each Run:

1. Launch LabRecorder.exe from the Start Menu.
2. In LabRecorder, go to **File**, then **Load Configuration**.
3. Choose the cfg file in emg directory (see 'Directories Shorthand' table).
4. Once loaded change the following:
    a. In **Block/Task**, change 'Default' to 'emg-grip'
    b. Update **Run** to whichever run of PsychoPy this corresponds to. Leave as 1 if your first run.
    c. Update **Participant**.
    d. Update **Session.** This will all be in session 2. Change to 02 instead of S001.
    e. Select 'beh' for **Modality.**

5. Next, Launch **PsychoPy Builder** from Start Menu.
6. In PsychoPy Builder, go to **File**, then **Open**, then navigate to the stim presentation directory and select the **grip_task.psyexp** file.
7. In PsychoPy Builder, switch the toggle in the top bar from **Pilot** to **Run**.
8. Move LabRecorder over to the second monitor.
9. Go back to MATLAB.
10. Navigate to the emg subdirectory (in UCBED directory).
11. Open the script recordEmgDataSyncWithPsychoPy.m.
12. Move MATLAB window over to the second monitor.
13. In PsychoPy, press the green **Run** button to start the experiment. Hit **OK**.
14. In the modal that pops up, input the participant (same ID as you entered in LabRecorder). Check that mode is 'run.'
15. In **paradigm**, update fmri to emg.
16. Ensure the conditions file number is correct. If first run, it's okay. If not first run, change to the appropriate filename.
17. Run the MATLAB script recordEmgDataSyncWithPsychoPy.m
18. Enter 0 (do not run in debug mode).
19. Type in participant ID (same as in PsychoPy and LabRecorder), for **Subject ID.**
20. Enter the **run number** (same as in PsychoPy and LabRecorder).
21. Run.
22. MATLAB will remind you to check for two green streams (one from Psychopy, one from MATLAB) in LabRecorder, and to press '**Start.**' If you don't see them, press the **Update** button first. Start Recording.
23. Continue to answer the MATLAB prompts till it says that it is okay to continue.
24. Move your cursor to the main monitor and press any key to skip instructions.
25. Press '**t'** to trigger the PsychoPy experiment.
26. At the end of the run, press **Stop** on LabRecorder.

    NOTE: At the end of a successful run, you should see:

    - in **subject-data/subID:** a file for each cam, repeat and grasp (5 grasps $\times$ 2 limbs $\times$ 2 types of files $\times$ 5 repeats = 100 mat files).
    - in **subject-data/sub-<SubID>/…**, an xdf file (with timing markers).

Recommended Checks After Run 1:

These can be done before the participant arrives with a brief test run on a lab member to check that everything works, if you'd rather save time.

a. Spot-check any RW movie using the helper function 'viewMovie(filepath)' (in 'Task_Peripheral' folder). It can be passed in the path to the mat file you want to view. Check to see the ultrasound shows movement, and the RW movie shows a full grasp, i.e., not blank, not stalled, and not misaligned in time.
b. Optionally, plot the channels of any of the EMG mat files to quickly view activity (OK if you don't).
c. Load the LSL (.xdf) file, with the function load_xdf(filepath_to_xdf) (make sure xdf-Matlab package in the Task_Peripheral folder is added to path).

   It should contain two non-empty structs (one for each stream).
   timings = load_xdf(filepath_xdf);
   timings{1}.time_series'
   timings{2}.time_series'

At the end of the full EMG Experiment:

1. Take off COBAN.
2. Take pictures of the set up/ location of sensors (with sensor numbers) on participant's arms.
3. Disconnect apparatus from participant.
4. Remove tapes from sensors, and put them back in Delsys placeholders.
5. Then pack them into Ziploc bag.
6. Remove power to Delsys.
7. Remove connections and pack up equipment.
8. Check that files were saved where you expect them to be saved.
9. Check roughly that there are the expected number of files (for 10 runs, 5 grasps per run times 5 repeats per run times 2 types of files times 2 limbs = 10.5.5.2.2 = 1000 files.
10. Spot-check a couple files with viewMovie.m before EMG session.

    e.g., for one trial, view the RW movie to see full grasp in unaffected side.

4. Troubleshooting

- Parallel pool error: The most common issue you may run into is if you exited the MATLAB program or it exited with errors and you attempt to rerun. It could say something like parallel pool processes can't be started. In that case, first kill existing pool by running in console: delete(gcp('nocreate')), which gets current pool and deletes it, so the script can create a new pool.

- <u>LSL error:</u> MATLAB may exit saying no LSL markers were found, or no stream was found. This happens if PsychoPy is not already on when MATLAB runs the scripts. If in doubt, clear clc MATLAB, reopen PsychoPy, and then rerun MATLAB script.
- <u>Directory './scratch' not found:</u> This can happen when you run the script from somewhere other than the directory it's in. Make sure you are in the ultrasound dir if running this exp, or the emg dir if running the EMG experiment.
- If there are any other issues, MATLAB console should say it has exited and name the exception. This should not happen during a data collection session, but if it does, the easiest way to debug would be to run again with lines 97(try), and 129-131 (catch, end block) commented out so that the full error message and line number can be displayed.

5. Running with One Screen

Because PsychoPy takes over the whole screen when it starts, it is not possible to press Start on LabRecorder if you open PsychoPy first. However, ideally before starting recording of LSL streams by LabRecorder, you should be able to check if both streams are recognized by LabRecorder before beginning the experiment, and the streams only come online when both PsychoPy and MATLAB scripts are started.

So there is a way to run everything with one monitor, but it requires bypassing the check that LabRecorder can see the streams.

1. You Open LabRecorder.

2. Load the config file for the experiment being run.

3. Change the fields as mentioned previously, in LabRecorder.

4. Then select all streams (PsychoPy Markers and one more – either emg or ultrasound depending on the experiment).

5. Press **Start Recording**. At the pop-up asking if you're sure, say **Yes** to record anyway.

6. Launch MATLAB.

7. Make sure PsychoPy is ready to run the grip_task.psyexp file, i.e., have it open.

8. Very quickly, launch MATLAB, enter the args it asks for, and then Run PsychoPy. It is possible MATLAB will timeout still. So Don't run with one monitor. If you must for testing, the go into the

MATLAB script and edit where it searches for LSL markers, and increase its waiting time so it doesn't time out. Then retry.